

10.2 DC: Potential Dividers

Question Paper

Course	CIE A Level Physics
Section	10. D.C. Circuits
Topic	10.2 DC: Potential Dividers
Difficulty	Hard

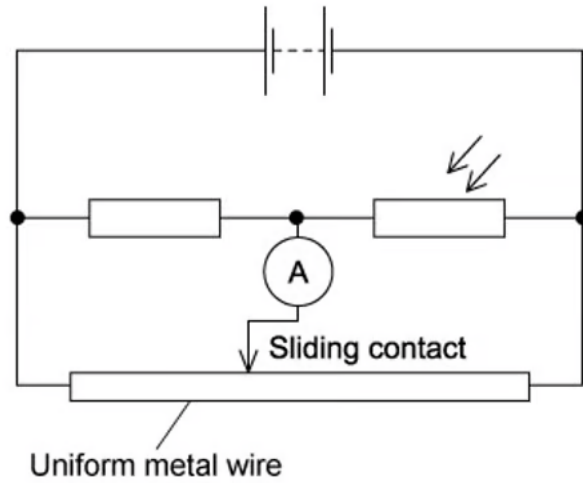
Time allowed: 20

Score: /10

Percentage: /100

Question 1

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered up, and the ammeter gives a non-zero reading.

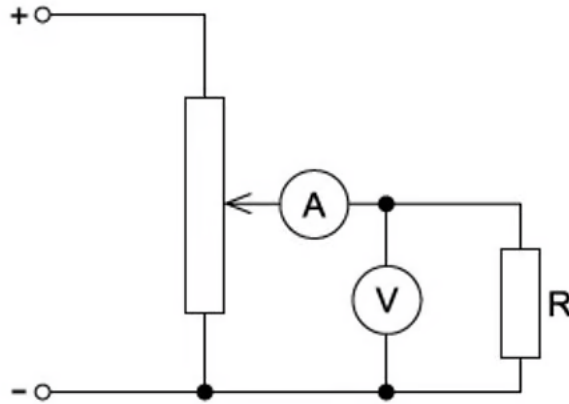
Which change could return the ammeter reading to zero?

- A. decrease the supply voltage
- B. increase the supply voltage
- C. move the sliding contact to the left
- D. move the sliding contact to the right

[1 mark]

Question 2

In the circuit below, a voltmeter of resistance R_V and an ammeter of resistance R_A are used to measure the resistance R of the fixed resistor.



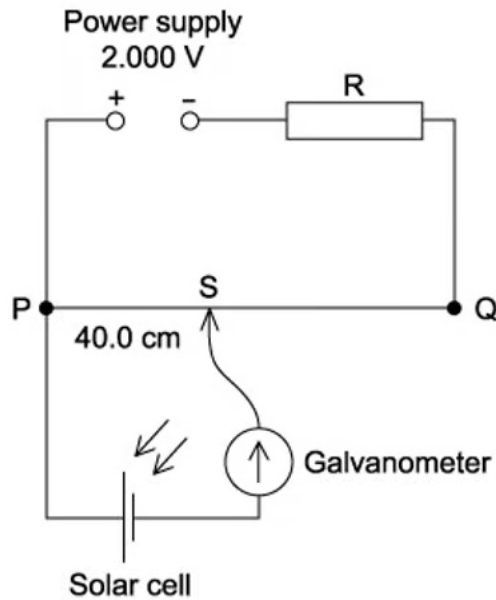
Which condition is necessary for an accurate value to be obtained for R ?

- A. R is much smaller than R_V
- B. R is much smaller than R_A
- C. R is much greater than R_V
- D. R is much greater than R_A

[1 mark]

Question 3

A power supply and a solar cell are compared using the potentiometer circuit shown.



The e.m.f. produced by the solar cell is measured on the potentiometer.

The potentiometer wire PQ is 100.0 cm long and has a resistance of $5.00\ \Omega$. The power supply has an e.m.f. of 2.000 V, and the solar cell has an e.m.f. of 5.00 mV.

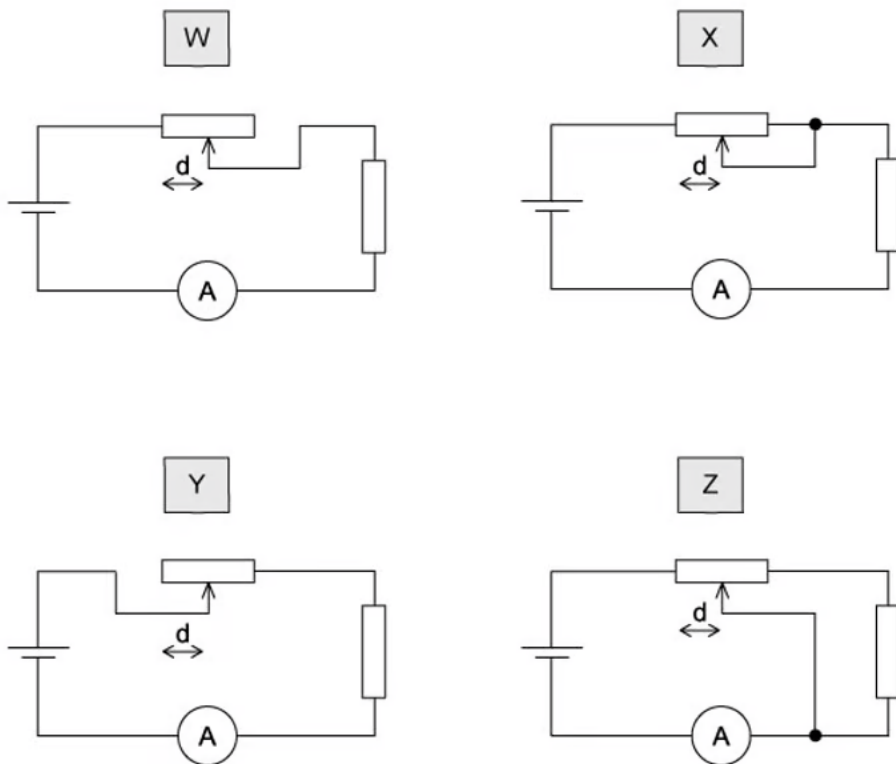
Which resistance R must be used so that the galvanometer reads zero when $PS = 40.0\text{ cm}$?

- A. $395\ \Omega$
- B. $795\ \Omega$
- C. $995\ \Omega$
- D. $1055\ \Omega$

[1 mark]

Question 4

The diagrams show the same cell, ammeter, potentiometer and fixed resistor connected in different ways.



The distance d between the sliding contact and a particular end of the potentiometer is varied. The current measured is then plotted against the distance d .

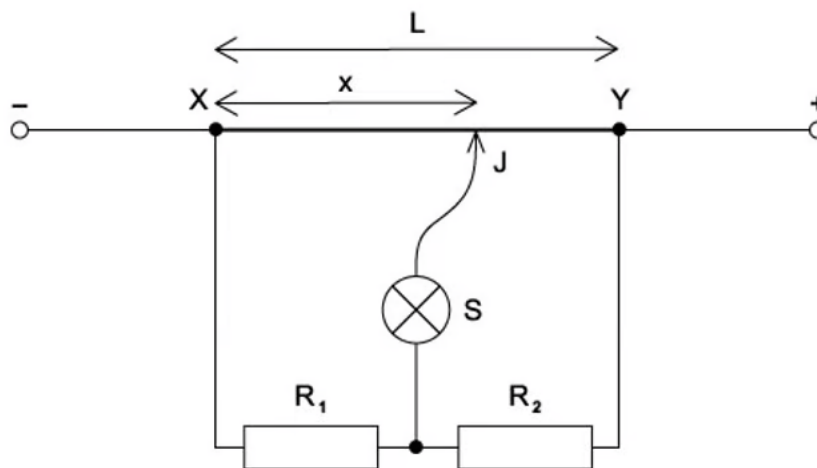
For which two circuits will the graphs be identical?

- A. W and X
- B. W and Y
- C. X and Y
- D. Y and Z

[1 mark]

Question 5

In the circuit shown, XY is a length L of uniform resistance wire. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a small signal lamp S.



To determine the ratio $\frac{V_1}{V_2}$ of the potential differences across R_1 and R_2 , a point is found on XY at which the lamp is off. This point is at a distance x from X.

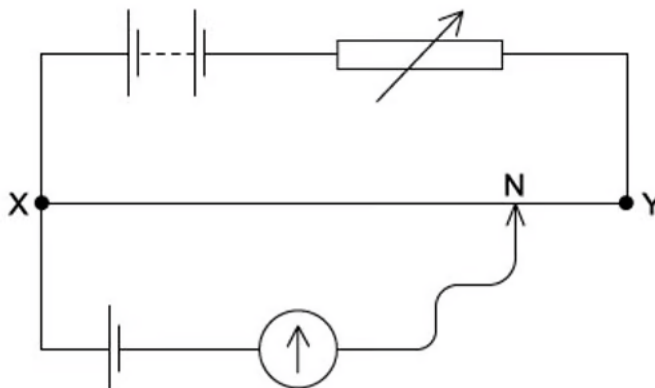
What is the value of the ratio $\frac{V_1}{V_2}$?

- A. $\frac{L}{x}$
- B. $\frac{x}{L}$
- C. $\frac{L-x}{x}$
- D. $\frac{x}{L-x}$

[1 mark]

Question 6

In the potentiometer circuit below, the moveable contact is placed at N on the bare wire XY, such that the galvanometer shows zero deflection.



The resistance of the variable resistor is now increased.

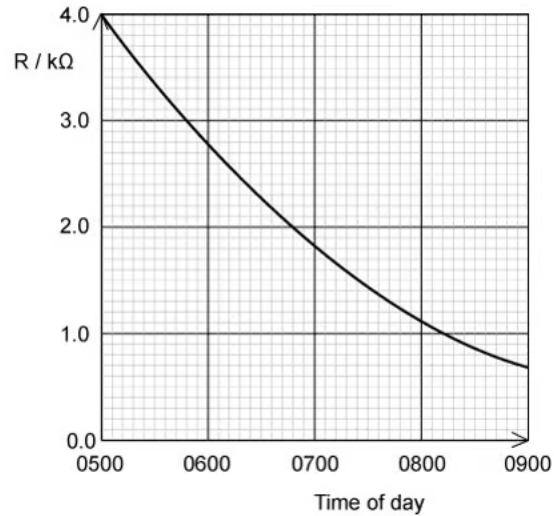
What is the effect of this increase on the potential difference across the wire XY and on the position of the moveable contact for zero deflection?

	Potential difference across XY	Position of moveable contact
A	increases	nearer to X
B	increases	nearer to Y
C	decreases	nearer to X
D	decreases	nearer to Y

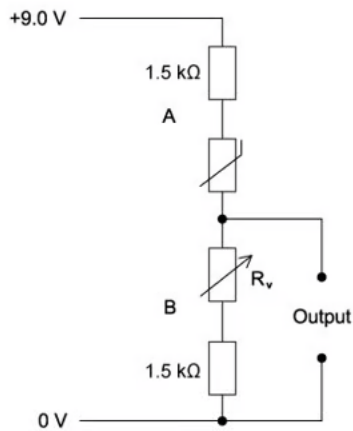
[1 mark]

Question 7

The graph below shows how resistance R of a thermistor varies during part of a day.



The thermistor is connected to the potential divider circuit as shown.



At 0730, an output of 6.0 V is required.

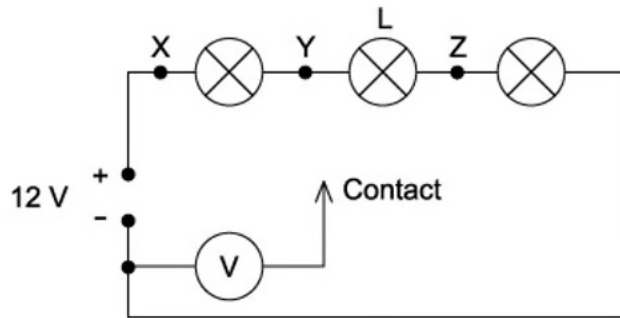
What value of the variable resistor should be set at this time?

- A. 1.5 kΩ
- B. 3.0 kΩ
- C. 4.5 kΩ
- D. 6.0 kΩ

[1 mark]

Question 8

The contact is connected one at a time to points X, Y and Z to test the circuit. The lamps do not light up because lamp L is blown.



Which of the following are the readings of the voltmeter when the contact is connected to each of the points?

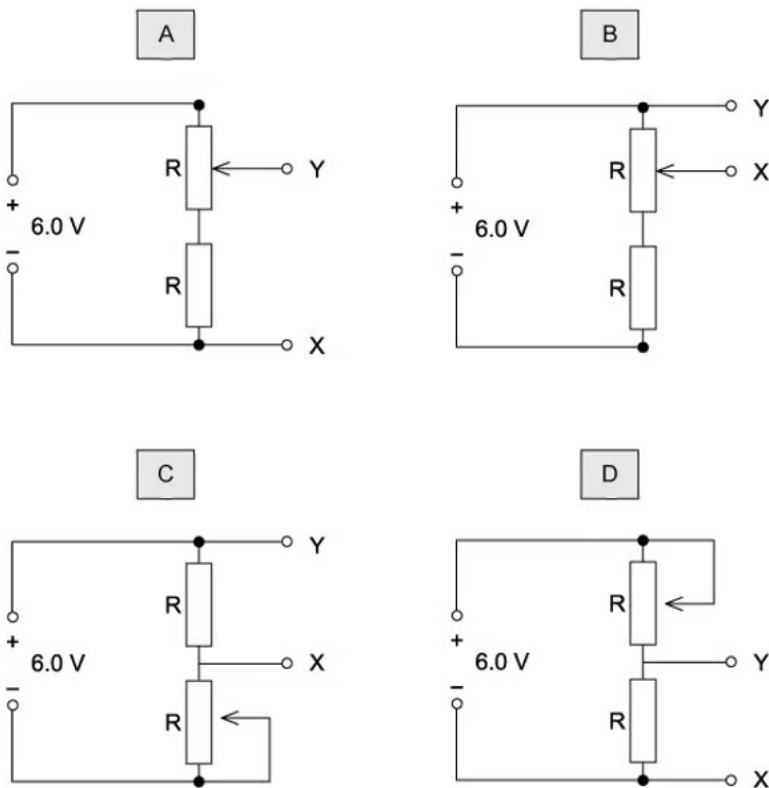
	X	Y	Z
A	12 V	8 V	4 V
B	8 V	8 V	0 V
C	12 V	12 V	0 V
D	8 V	12 V	4 V

[1 mark]

Question 9

A potential divider has a constant supply of 6.0 V , as shown in the diagrams.

Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0 V ?

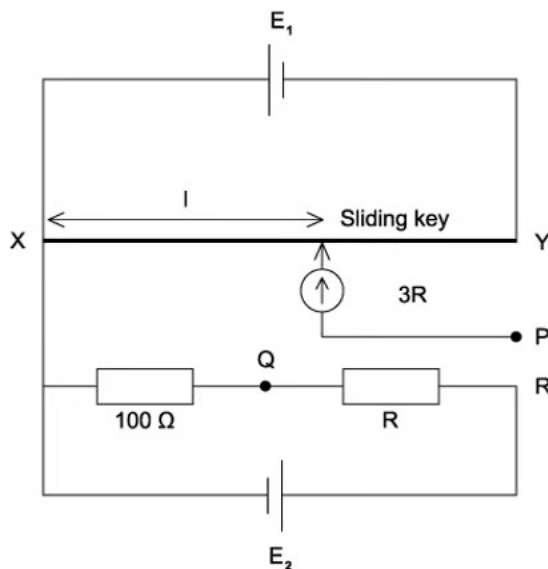


[1 mark]

Question 10

The resistance R of an unknown resistor is compared with a fixed $100\ \Omega$ resistor using the circuit as shown in the diagram.

The balance length l is 30.0 cm and 48.3 cm when P is connected to Q and R , respectively.



The length of the wire XY is 100 cm and has a resistance of $40\ \Omega$. The two cells E_1 and E_2 have negligible internal resistances.

What is the value of resistance R ?

- A. $38\ \Omega$
- B. $47\ \Omega$
- C. $61\ \Omega$
- D. $161\ \Omega$

[1 mark]



Model Answers: Hard

1

The correct answer is **C** because:

- For the ammeter to read zero, the potential difference across it should be zero; hence, the terminals of the ammeter are at the same potential
- The resistance of an LDR decreases when the light intensity on it increases, so, as the LDR is covered up, the light intensity decreases – this causes its resistance to increase
- Considering the resistor and the LDR in the middle section, an increase in the resistance of the LDR causes the p. d. across it to increase ($V = IR$)
 - So, the terminal connected between the LDR and the resistor is now at a higher potential than before
- To return the ammeter reading to zero, the sliding contact should be moved in such a way that it is at a higher potential (and equal to the new potential of the other terminal of the ammeter)
- The right end of the uniform metal wire is connected to the negative terminal of the supply and so, is at a potential of zero
- The left end of the wire is connected to the positive terminal of the supply and is at the maximum potential (equal to the e. m. f of the supply)
- So, as we move from right to left, the potential increases
 - Therefore, to increase the potential of the sliding contact, we need to move it to the left

2

The correct answer is **A** because:

- To obtain the resistance of resistor R from the set-up shown, the potential difference and current flowing through the resistor are measured
 - Ohm's law states $V = IR$, so resistance $R = \frac{V}{I}$
- Ideally, a voltmeter should have an infinite resistance and be connected in parallel to the component being investigated – this is to avoid a large current

flowing through it since current is highest when the resistance is smallest

- Ideally, an ammeter should have negligible resistance and be connected in series to the component being investigated so that it does not affect the value of the current flowing
- So, R_V should be very large (larger than R), and R_A should be very small (smaller than R).
 - Therefore, in order to obtain an accurate value of R , the necessary condition is that R should be much smaller than R_V

B & D are incorrect because the ammeter is in series with R , so the ammeter will always measure the current through R , no matter what the value of R or R_A . If R_A is either small or large, both V and I will change proportionally $\left(R = \frac{V}{I}\right)$ and R won't change

C is incorrect because an ideal voltmeter has infinite resistance, so R could not be higher than this

3

The correct answer is **B** because:

- A galvanometer is a type of sensitive ammeter an instrument for detecting electric current
- In order for the galvanometer to read zero, the e.m.f of the solar cell must be equal to the p.d. such that the p.d. across the galvanometer is zero
- The potential divider equation is given by
 - $V_{out} = V_0 \left(\frac{R_1}{R_1 + R_2} \right)$
- When $PS = 40.0$ cm, the p.d. across the length of wire, PS is given by
 - $2.000 \text{ V} \times \left(\frac{0.400 \times 5.00}{R + 5.00} \right)$
- So, by equating this p.d. to the e.m.f. of solar cell (5.00 mV), resistance R can be obtained by rearranging the equation:
 - $2.000 \times \left(\frac{0.400 \times 5.00}{R + 5.00} \right) = 5.0 \times 10^{-3} \text{ V}$
 - $2.000 \times 0.400 \times 5.00 = (5.0 \times 10^{-3})(R + 500)$
 - $4.00 = (5.0 \times 10^{-3})R + 0.025$

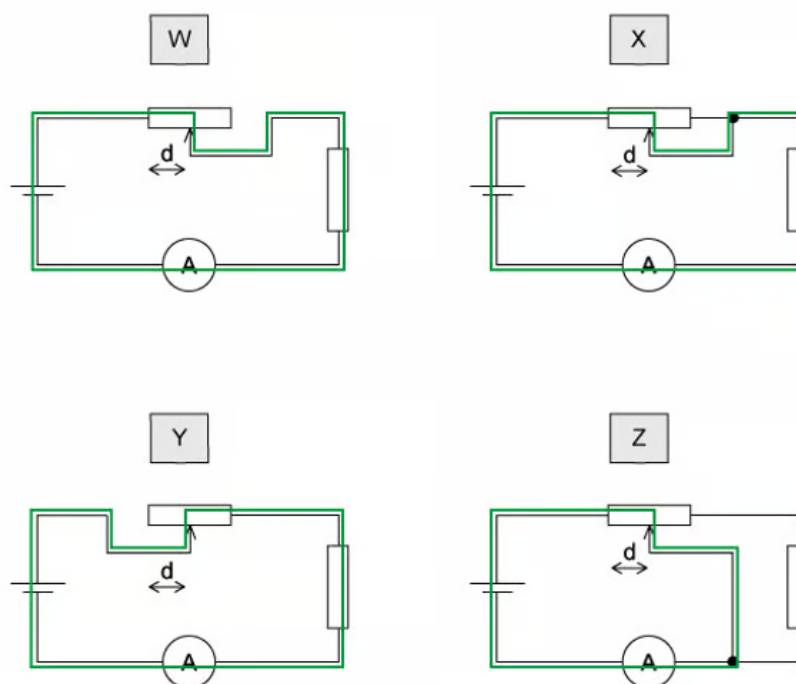
$$\circ R = \frac{4.00 - 0.025}{5.0 \times 10^{-3}} = \frac{3.975}{5.0 \times 10^{-3}} = 795 \, \Omega$$

- Therefore, the resistance R is equal to $795 \, \Omega$

4

The correct answer is **A** because:

- A potentiometer is different than a regular resistor in that the potentiometer can allow for different values of resistance
- Current always chooses the easiest way, that is, the path is the smallest resistance
- The best way to approach this question is to draw out the path of the current in each circuit
- As we can see from the diagram, the path of the current is the same in W and X; therefore, they will give identical graphs



B & C are incorrect because in arrangement Y, the path other than the distance d on the wire of the potentiometer is connected to the circuit. So, this arrangement is different from those of W and X. Even if the circuit is complete, the results obtained on a graph would not be identical to that of W or X

D is incorrect because in arrangement Z, current flows from the positive terminal of the battery, then along a distance d on the potentiometer, but here, the current does not flow through the fixed resistor; hence it would not give the same graph as in Y

5

The correct answer is **D** because:

- For the lamp to be off, the potential difference across it should be zero so that no current flows through it
 - This means, the potential at the junction of R_1 and R_2 should be equal to the potential at the sliding contact J
- The p. d. across R_1 is V_1 and the p. d. across R_2 is V_2 , so, V_1 should be equal to the p. d. across XJ and V_2 should be equal to the p. d. across JY
- Since point X is connected to the negative terminal of the supply, it can be considered to be at a potential of 0, while point Y is at the maximum potential, equal to the e.m.f. of the supply
- The wire XY, together with the sliding contact J, can be considered as a potential divider
- Resistance R of the wire $= \frac{\rho L}{A}$ and from Ohm's Law $V = IR$, so, the p. d. across the wire is proportional to the length of wire $V \propto L$
- A length of L corresponds to a p. d. equal to the e.m.f. E of the supply and a length of x corresponds to a p. d. equal of $\left(\frac{E}{L}\right) \times x = \left(\frac{x}{L}\right) \times E$
- Since X is at a potential of zero, point J would be at a potential of:
 - $V_1 = \left[\left(\frac{x}{L}\right) \times E\right] - 0 = \left(\frac{x}{L}\right) \times E$
- Now, the part JY of the wire corresponds to a length $(L - x)$ of the wire. A length of $(L - x)$ corresponds to a p.d. equal to
 - $V_2 = \left(\frac{E}{L}\right) \times (L - x) = \frac{(L - x)}{L} \times E$
- Therefore, the ratio $= \frac{V_1}{V_2} = \left(\frac{x}{L}\right) E \div \frac{(L - x)}{L} E = \frac{x}{L - x}$

5



The correct answer is **D** because:

- For the lamp to be off, the potential difference across it should be zero so that no current flows through it
 - This means, the potential at the junction of R_1 and R_2 should be equal to the potential at the sliding contact J
- The p. d. across R_1 is V_1 and the p. d. across R_2 is V_2 , so, V_1 should be equal to the p. d. across XJ and V_2 should be equal to the p. d. across JY
- Since point X is connected to the negative terminal of the supply, it can be considered to be at a potential of 0, while point Y is at the maximum potential, equal to the e. m. f. of the supply
- The wire XY, together with the sliding contact J, can be considered as a potential divider
- Resistance R of the wire $= \frac{\rho L}{A}$ and from Ohm's Law $V = IR$, so, the p. d. across the wire is proportional to the length of wire $V \propto L$
- A length of L corresponds to a p. d. equal to the e. m. f. E of the supply and a length of x corresponds to a p. d. equal of $\left(\frac{E}{L}\right) \times x = \left(\frac{x}{L}\right) \times E$
- Since X is at a potential of zero, point J would be at a potential of:
 - $V_1 = \left[\left(\frac{x}{L}\right) \times E\right] - 0 = \left(\frac{x}{L}\right) \times E$
- Now, the part JY of the wire corresponds to a length $(L - x)$ of the wire. A length of $(L - x)$ corresponds to a p. d. equal to
 - $V_2 = \left(\frac{E}{L}\right) \times (L - x) = \frac{(L - x)}{L} \times E$
- Therefore, the ratio $= \frac{V_1}{V_2} = \left(\frac{x}{L}\right) E \div \frac{(L - x)}{L} E = \frac{x}{L - x}$

6

The correct answer is **D** because:

- Let's say the resistance of the variable resistor $= R$
 - e. m. f. of driver cell $=$ p. d. across variable resistor $+$ p. d. across wire XY
- As the resistance R of the variable resistor is increased, the p. d. across it also

increases ($V = IR$) and hence, the p. d. across the wire XY decreases

- For the galvanometer to show zero deflection, the potential difference across it should be zero, i.e. no current flowing through it
 - This means the potential at one side of the galvanometer should be equal to the potential on the other side, making the potential difference zero
- Initially, before the resistance of the variable resistor is increased, the galvanometer showed zero deflection
 - So, the potential difference across it was zero and the p. d. across XN was equal to the e.m.f. of the cell
- But an increase in the resistance of the variable resistor causes the p. d. across XY to decrease
 - So, to cause the galvanometer to show zero deflection again, the p. d. across wire XN should be increased so that it becomes equal to the e.m.f. of the cell again
- Using the equations $V = IR$ and $R = \frac{\rho L}{A}$, we can see that $V \propto L$
 - Therefore, to increase the p. d. the length must be increased
 - This is done by moving the contact closer to end Y

A & B are incorrect because when the resistance R increases, the p. d. across it also increases and hence, the p. d. across the wire XY decreases

C is incorrect because to cause the galvanometer to show zero deflection again, the moveable contact must move nearer to Y since $V \propto L$

7

The correct answer is **C** because:

- Section A is made up of a $1.5 \text{ k}\Omega$ fixed resistor and the thermistor
- Section B is made up of the variable resistor and another $1.5 \text{ k}\Omega$ fixed resistor
 - If the output is to be set at 6.0 V , then across the section A, the voltage drop needs to be 3.0 V
- From the graph, at 0730 , the resistance of the thermistor = $1.5 \text{ k}\Omega$
- From Ohm's Law: $V = IR$
- The current flowing through section A:
 - $I = \frac{V}{R} = \frac{3.0 \text{ V}}{(1500 + 1500)} = 0.001 \text{ A}$
- Similarly, the same current flows through section B:

- $6.0 \text{ V} = IR = (0.001)(R_v + 1500)$
- $0.001 R_v = 6.0 - (1500 \times 0.001) = 4.5$
- $R_v = \frac{4.5}{0.001} = 4500 \Omega = 4.5 \text{ k}\Omega$
- Therefore, at 0730 the variable resistor should be set to $4.5 \text{ k}\Omega$

8

The correct answer is **C** because:

- For point X, the voltmeter is merely measuring the voltage between the terminals of the battery as there are no electrical components in between
 - Hence, the reading is that of the voltage of the battery, which is 12 V
- For point Y, there is a lamp in between the ends of the voltmeter
- However, due to the broken filament of lamp L, no current flows in the circuit
- According to $V = IR$, the voltage across the lamp between X and Y will be zero
 - Hence, the voltage measured by the voltmeter will still be that between the terminals of the battery, which is 12 V
- Finally, placing the contact at point Z has the same effect as placing it at the negative terminal, so, the voltmeter will read the voltage between the negative terminal and itself
 - Hence, the reading registered will be 0 V

9

The correct answer is **B** because:

- Since the two resistors are in series, the total e.m.f in the circuit is divided between them
- The variable resistor controls the range of the output voltage, so X and Y must be placed across the variable resistor
- Consider option B:
 - When the resistance of the top resistor is maximized to R , the potential difference between X and Y = 3.0 V
 - When the resistance of the top resistor is minimized to 0 , the potential difference between X and Y = 0 V
- Therefore, option **B** is correct

A is incorrect because the range of output voltage of this circuit would be 3 – 6 V as it includes both resistors

C & D are incorrect because the variable resistor controls the range of the output voltage, so X and Y must be placed across the variable resistor

10

The correct answer is **C** because:

- When P is connected to Q, the potential difference across the balance length is equal to the potential difference across the 100 Ω resistor

$$\frac{l}{XY} \times E_1 = \frac{100}{100 + R} \times E_2 \quad (1)$$

- When P is connected to R, the potential difference across the balance length is equal to the potential difference across the 100 Ω resistor and the unknown resistor (which is the same as the e. m. f. as cell E)

$$\frac{l'}{XY} \times E_1 = E_2 \quad (2)$$

- Substitute equation (2) into (1)

$$\frac{l}{XY} \times E_1 = \frac{100}{100 + R} \times \frac{l'}{XY} \times E_1$$

$$\frac{30.0}{XY} \times E_1 = \frac{100}{100 + R} \times \frac{48.3}{XY} \times E_1$$

- The E_1 and XY terms cancel out, so we have:

$$30.0 = \frac{100}{100 + R} \times 48.3$$

$$100 + R = \frac{100 \times 48.3}{30.0} = 161$$

$$R = 161 - 100 = 61 \Omega$$

- Therefore, the resistance R of the unknown resistor is 61 Ω

